Kubernetes ReplicaSet Practical

Step-by-step Minikube lab • Commands with uses • YAML configuration • Verification • Viva

Objective: Start a local Kubernetes cluster, create a ReplicaSet that maintains three NGINX Pods, inspect the resources, delete one Pod, and observe automatic replacement.

1. What you will build

Minikube runs a local Kubernetes cluster. A ReplicaSet named **nginx-replicaset** makes sure that three Pods matching the label `app: nginx` exist. Each Pod runs an NGINX container.

Component	Purpose
Minikube	Runs a local Kubernetes cluster on your computer.
kubectl	Command-line tool to communicate with Kubernetes.
ReplicaSet	Maintains the requested number of matching Pods.
Pod	Smallest deployable Kubernetes unit; contains one or more containers.
NGINX image	Container image used by each Pod; NGINX listens on port 80 inside the container.

2. Start and check Minikube

Step 1 — Start the cluster

```
minikube start
```

Use: Starts the local Kubernetes cluster. Wait until the command reports that Kubernetes has started.

Step 2 — Check the node

```
kubectl get nodes
```

Use: Lists the Kubernetes nodes. The Minikube node should normally show `Ready`.

Step 3 — Check Minikube components

```
minikube status
```

Use: Checks whether Minikube components such as the host, kubelet, and API server are running.

If a command is not recognized: Install the missing tool, reopen the terminal, and retry. Minikube also needs a supported container runtime or virtualization driver.

3. Create the working folder

In PowerShell, go to a location where you want to store your lab files. These commands use your Desktop as an example:

```
cd $HOME\Desktop
mkdir kubernetes-replica
cd kubernetes-replica
pwd
```

Uses: Moves to the Desktop, creates the project folder, enters it, and prints the current working directory. If the folder already exists, simply use `cd $HOME\Desktop\kubernetes-replica`.

4. Create the YAML configuration

```
notepad replicaset.yaml
```

Use: Opens Notepad to create the Kubernetes manifest. If asked, confirm creation of the new file.

Paste the following YAML exactly, then save with Ctrl+S. Ensure the filename is `replicaset.yaml`, not `replicaset.yaml.txt`.

```
apiVersion: apps/v1
kind: ReplicaSet

metadata:
  name: nginx-replicaset

spec:
  replicas: 3

  selector:
    matchLabels:
      app: nginx

  template:
    metadata:
      labels:
        app: nginx

    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

What the important YAML fields mean

Field	Meaning
apiVersion: apps/v1	Kubernetes API group/version for ReplicaSet.
kind: ReplicaSet	Declares the resource type.
metadata.name	Names the ReplicaSet nginx-replicaset.
replicas: 3	Requests three Pod replicas.
selector.matchLabels	Selects Pods with the matching label.
template	Describes how Kubernetes should create new Pods.
image: nginx:latest	Uses the NGINX container image.
containerPort: 80	Documents the port used inside the container; it does not expose the app externally by itself.

Important: The Pod template label `app: nginx` must match the ReplicaSet selector `app: nginx`.

5. Apply the configuration

```
kubectl apply -f replicaset.yaml
```

Use: Creates or updates the resource described in the YAML file. Expected first-time result: `replicaset.apps/nginx-replicaset created`. If already applied without changes, it may report `unchanged`.

6. Verify the ReplicaSet and Pods

Check the ReplicaSet

```
kubectl get replicasets
```

Use: Lists ReplicaSets. Look for `nginx-replicaset` and values of 3 for DESIRED, CURRENT, and eventually READY.

DESIRED = requested Pod count; **CURRENT** = existing matching Pods; **READY** = Pods ready to serve.

Check Pods

```
kubectl get pods
```

Use: Lists Pods in the current namespace. Expect three Pod names beginning with `nginx-replicaset-`. Each should eventually show `1/1 Running`.

If a Pod temporarily shows `ContainerCreating`, wait a short time and check again while the image downloads.

Describe the ReplicaSet

```
kubectl describe replicaset nginx-replicaset
```

Use: Displays detailed information including selector, replica counts, Pod template, and Events. Check Events for creation messages or errors.

List common resources

```
kubectl get all
```

Use: Shows common resource types in the current namespace, including the ReplicaSet and Pods. It does not literally show every Kubernetes resource type.

7. Test self-healing by deleting one Pod

First, list the Pods and copy one actual Pod name:

```
kubectl get pods
```

Delete one Pod using its exact name. Replace the example placeholder below with a real name from your output:

```
kubectl delete pod <actual-pod-name>
```

Use: Deletes the selected Pod. Do not type the angle-bracket placeholder literally; use the full Pod name shown by `kubectl get pods`.

Now check again:

```
kubectl get pods
kubectl get replicasets
```

A new Pod should be created to restore the desired count of three. Its name will normally differ from the deleted Pod's name. The replacement may take a little time to become Ready.

Why? The ReplicaSet continually compares the desired state (three Pods) with the actual state. When one is missing, it creates a replacement. This is called reconciliation/self-healing. The replacement is a new Pod instance, not the same Pod restored.

8. Full command checklist

Purpose	Command
Start Minikube	`minikube start`
List nodes	`kubectl get nodes`
Check Minikube status	`minikube status`
Create folder (PowerShell example)	`mkdir kubernetes-replica`
Enter folder	`cd kubernetes-replica`
Create/edit YAML	`notepad replicaset.yaml`
Apply manifest	`kubectl apply -f replicaset.yaml`

Purpose	Command
List ReplicaSets	`kubectl get replicasets`
List Pods	`kubectl get pods`
Show ReplicaSet details	`kubectl describe replicaset nginx-replicaset`
List common resources	`kubectl get all`
Delete one Pod	`kubectl delete pod <actual-pod-name>`
Verify Pod replacement	`kubectl get pods`

9. Clean up after the practical

```
kubectl delete -f replicaset.yaml
kubectl get pods
minikube stop
```

Uses: Deletes the ReplicaSet and its managed Pods, checks the remaining Pods, and stops Minikube. Run `minikube stop` only when you have finished using the local cluster.

10. Quick viva questions

Q. What is Kubernetes?
An open-source container orchestration platform used to deploy, manage, scale, and recover containerized applications.

Q. What is Minikube?
A tool for running a local Kubernetes cluster, useful for learning and testing.

Q. What is a Pod?
The smallest deployable unit in Kubernetes, containing one or more containers.

Q. What is a ReplicaSet?
A Kubernetes resource that maintains a specified number of matching Pod replicas.

Q. Why did a new Pod appear after deletion?
The ReplicaSet detected fewer Pods than desired and created a replacement.

Q. What is the difference between desired and current replicas?
Desired is the requested count; current is the count that currently exists.

Q. Does a ReplicaSet expose NGINX to a browser?
No. A Service or another networking configuration is generally needed to expose the application.

Expected outcome: Three NGINX Pods managed by one ReplicaSet. After deleting one Pod, Kubernetes creates a replacement so the ReplicaSet returns to three Pods.

Troubleshooting tip: If Pods do not become Ready, run `kubectl describe pod <actual-pod-name>` and inspect the Events section. Check internet/image-pull access and the Minikube cluster status.